Table 4.2 Properties of water, heavy water and super heavy water†.

Property	H_2O	D_2O	T_2O
Molecular weight	18.015	20.0276	22.031
Melting point (*K*)	273.0	276.8	277.5
Boiling point (*K*)	373.0	374.4	374.5
Temperature of maximum density (*K*)	277.0	284.2	286.4
Maximum density (g/cm^3)	1.000	1.106	1.215
Density (g/cm^3)	0.997	1.104	1.214
Vapour pressure (*mm* Hg)	23.75	20.51	19.80
Viscosity (*cP*)	0.890	1.107	---
Dielectric constant	78.39	78.06	---
Enthalpy of formation (*kJ/mol*)	– 285.9	– 294.6	---
Enthalpy of vaporization (*kJ/mol*)	40.66	41.61	---

†Unless otherwise stated, all data are at 298 *K*.

4.5.3 Chemical Properties:

Water reacts with metals, non-metals and other compounds differently. The most reactive metals are the alkali metals. They decompose water even in cold with the evolution of hydrogen leaving an alkali solution.

$$2Na + 2\ H_2O \rightarrow 2\ NaOH + H_2$$

The group 2 metals (except beryllium) react in a similar way but less violently. The hydroxides are less soluble than those of Group 1.

$$Ba + 2H_2O \rightarrow Ba(OH)_2 + H_2$$

Some transition metals react with hot water or steam to form the corresponding oxides. For example, steam passed over red hot iron results in the formation of iron oxide with the release of hydrogen.

$$3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$$

Lead and copper decompose water only at a white heat. Silver, gold, mercury and platinum do not have any effect on water. In the elemental form, the non-metals such as carbon, sulphur and phosphorus normally do not react with water. However, as we have seen earlier, carbon will react with steam when it is red (or white) hot to give *water gas*.

11th Std Chemistry 100-124.indd 107 06-01-2021 15:24:55